

Wei-En Hsieh

ann20010929@gmail.com — LinkedIn — GitHub — Portfolio

EDUCATION

Swiss Federal Institute of Technology in Zurich (ETH Zürich)

Master of Science in Cyber Security, Minor in Data Management Systems

Sep 2024 - Expected Feb 2028

📍 Zurich, Switzerland

- Relevant Coursework: Advanced Systems Lab, Cloud Computing Architecture, Security Engineering

Swiss Federal Institute of Technology in Lausanne (EPFL)

Bachelor of Science in Computer Science

Sep 2021 - Jul 2024

📍 Lausanne, Switzerland

- Relevant Coursework: Algorithms, Operating Systems, Parallelism and Concurrency, System Programming

WORK EXPERIENCE

ETH Zürich — Information Security Group

Incoming Research Assistant Python

Oct 2026 - Feb 2027

📍 Zurich, Switzerland

- Selected to build an LLM-powered pipeline for **Python instrumentation and security-policy verification**.

Bouygues Telecom

Automation Project Manager (Software Engineering) Python, Camunda, BPMN, REST APIs

Mar 2026 - Sep 2026

📍 Paris, France

- Automated an end-to-end Camunda workflow handling **30–50 mobile-network complaints/day**, leaving only 2 manual exception paths.
- Integrated **4 APIs (15 calls/complaint)** with retries, reusable components, and Gemini-based report analysis.
- Owned delivery from telecom expert requirements to production, translating them into validated BPMN workflows.

WasteFlow

Full-Stack Developer TypeScript, React, Node.js, SQL

Jul 2023 - Jun 2024

📍 Lausanne, Switzerland

- Built an operations platform for **25+ recycling-factory machines**, surfacing machine failures to operators.
- Partnered with 10+ domain experts to refine requirements and validate the platform against factory workflows.
- Implemented interactive dashboards to analyze machine performance and identify operational patterns.

EPFL

Teaching Assistant — Human-Computer Interaction (HCI), Linear Algebra, Mechanics

Sep 2022 - Jun 2024

📍 Lausanne, Switzerland

- Mentored 40+ HCI students building web and mobile UI/UX prototypes, reviewing functionality, usability, and implementation.
- Taught and supported students in Linear Algebra and Mechanics through exercises and problem solving.

RESEARCH & ENGINEERING PROJECTS

GDPRFS: FUSE-Based GDPR-Compliant File System Python, FUSE, Flask, SQLAlchemy

Sep 2025 - Apr 2026

- Built GDPRFS to enforce runtime policies without modifying applications that access files.
- Delivered **0–80 ms** enforcement overhead **across 100–10,000 files**, excluding LLM inference.
- Implemented page and row redaction on revocation and Lex mappings from FUSE events to GDPR concepts.
- Co-authored “Lex: Turning Laws into Enforceable Security Policies” accepted at **ACM CCS 2026**, and contributed the GDPRFS case study.

High-Performance Parallel Algorithm for Texture Synthesis C, AVX, Linux, perf

Feb 2025 - Jun 2025

- Optimized the algorithm with AVX intrinsics on Linux, achieving **13× speedup and 7+ ops/cycle** on CPU.
- Automated profiling and benchmarking with Linux perf to identify CPU bottlenecks.

WanderPals: Collaborative Android Travel Platform Kotlin, Jetpack Compose, Firebase

Feb 2024 - Jun 2024

- Implemented real-time suggestion search, sorting and role-based 24-hour voting with state persisted in Firestore.
- Built majority-based itinerary updates and agenda state tracking with **93%+ new-code coverage**.

LEADERSHIP & VOLUNTEERING

Paris Organising Committee for the 2024 Olympic and Paralympic Games

Volunteer, Sport Info & Administration

Jul 2024 - Sep 2024

📍 Paris, France

- Supported athlete operations for Marathon and Table Tennis.
- Escorted medalists to the stage and coordinated on-time podium ceremonies.

SKILLS

Programming

Python, C, C++, Java, JavaScript, TypeScript, Kotlin, SQL

Technologies

Linux/Unix, Docker, Git, GitHub Actions, Bash, React.js, Node.js, Flask, MySQL, Firebase

Languages

French (bilingual), Mandarin (native), English (professional), German (A2)